

Lecture-19
Mathematics 1 (15B11MA111)

CO [C105.4]

Module: Vector Differentiation

Topics to be covered

- **Divergence**

 - **Examples**

- **Curl**

 - **Examples**

- **Assignments**

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Divergence of a vector field

Let $\vec{v} = v_1(x, y, z)\hat{i} + v_2(x, y, z)\hat{j} + v_3(x, y, z)\hat{k}$ be a vector field, then the divergence of this vector field is defined as

$$\begin{aligned} \text{div } \vec{v} &= \nabla \cdot \vec{v} = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}) \\ &= \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z} \end{aligned}$$

Example 1: Find the divergence of the vector field $\vec{v} = (x^2y^2 - z^3)\hat{i} + 2xyz\hat{j} + e^{xyz}\hat{k}$

$$\begin{aligned} \text{Solution: } \text{div } \vec{v} &= \nabla \cdot \vec{v} = \frac{\partial(x^2y^2 - z^3)}{\partial x} + \frac{\partial(2xyz)}{\partial y} + \frac{\partial(e^{xyz})}{\partial z} \\ &= 2xy^2 + 2xz + xye^{xyz} \end{aligned}$$

Example2: if $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$, show that $\text{div} \left(\frac{\vec{r}}{r^3} \right) = 0$.

Solution: We have, $\nabla \cdot \left(\frac{\vec{r}}{r^3} \right) = \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot \left(\hat{i} \frac{x}{r^3} + \hat{j} \frac{y}{r^3} + \hat{k} \frac{z}{r^3} \right)$

$$= \left(\frac{\partial}{\partial x} \left(\frac{x}{r^3} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r^3} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r^3} \right) \right) = \frac{3}{r^3} - \frac{3}{r^4} \left(x \frac{\partial r}{\partial x} + y \frac{\partial r}{\partial y} + z \frac{\partial r}{\partial z} \right)$$

Since $r^2 = x^2 + y^2 + z^2$, we get $\frac{\partial r}{\partial x} = \frac{x}{r}$, $\frac{\partial r}{\partial y} = \frac{y}{r}$ and $\frac{\partial r}{\partial z} = \frac{z}{r}$.

Therefore,

$$\nabla \cdot \left(\frac{\vec{r}}{r^3} \right) = \frac{3}{r^3} - \frac{3}{r^5} (x^2 + y^2 + z^2) = 0$$

Example 3: Find the divergence of the vector field $\vec{v} = (xyz)\hat{i} + 2x^2y\hat{j} + (xz^2 - y^2z)\hat{k}$.

Solution: $\operatorname{div} \vec{v} = \nabla \cdot \vec{v} = \frac{\partial(xyz)}{\partial x} + \frac{\partial(2x^2y)}{\partial y} + \frac{\partial(xz^2 - y^2z)}{\partial z}$
 $= yz + 2x^2 + 2xz - y^2$

Curl of a vector field

Let $\vec{v} = v_1(x, y, z)\hat{i} + v_2(x, y, z)\hat{j} + v_3(x, y, z)\hat{k}$ be a vector field, then the curl of this vector field is given as

$$\operatorname{curl} \vec{v} = \nabla \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_1 & v_2 & v_3 \end{vmatrix} = \sum \left(\frac{\partial v_3}{\partial y} - \frac{\partial v_2}{\partial z} \right) \hat{i}$$

Example 4: Find the curl of a vector field $\vec{v} = (x^2y^2 - z^3)\hat{i} + 2xyz\hat{j} + e^{xyz}\hat{k}$.

Solution: $\text{curl } \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y^2 - z^3 & 2xyz & e^{xyz} \end{vmatrix}$

$$= \hat{i}(xze^{xyz} - 2xy) - (yze^{xyz} + 3z^2)\hat{j} + (2yz - 2x^2y)\hat{k}$$

Remarks:

1. Let $\vec{v}$ represents the velocity of a fluid in a medium and if $\text{div}(\vec{v})=0$, then the fluid is **incompressible**.
2. In electromagnetic theory, if $\text{div}(\vec{v})=0$, then the vector $\vec{v}$ is said to be **solenoidal**.
3. Rotation is also used in place of curl. In fluid mechanics, $\text{curl}(\vec{v})$ where $\vec{v}$ is the velocity, measures the vorticity of the fluid. If $\text{curl}(\vec{v})=0$ then $\vec{v}$ is said to be **irrotational**.
4. A force field $\vec{F}$ is said to be **conservative** if it is derivable from a potential function f , that is $\vec{F} = \text{grad}(f)$. Therefore, if $\vec{F}$ is conservative then $\text{curl}(\vec{F})=0$ and there exists a scalar potential function f such that $\vec{F} = \text{grad}(f)$.

Example 3. Show that the vector $\bar{v} = e^{x+y-2z}(\hat{i} + \hat{j} + \hat{k})$ is solenoidal.

Solution: Since $\bar{v} = e^{x+y-2z}(\hat{i} + \hat{j} + \hat{k})$

The divergence of $\bar{v}$ is given as

$$\begin{aligned} \text{div } \bar{v} &= \nabla \cdot \bar{v} = \frac{\partial(e^{x+y-2z})}{\partial x} + \frac{\partial(e^{x+y-2z})}{\partial y} + \frac{\partial(e^{x+y-2z})}{\partial z} \\ &= e^{x+y-2z} + e^{x+y-2z} - 2e^{x+y-2z} = 0 \end{aligned}$$

The $\text{div } \bar{v} = 0$, therefore $\bar{v}$ is solenoidal.

Example 4. Show that the vector $\bar{v} = e^{xy}(y\hat{i} + x\hat{j}) + 2e^z\hat{k}$ is irrotational.

Solution: $\bar{v} = e^{xy}(y\hat{i} + x\hat{j}) + 2e^z\hat{k}$.

The vector $\bar{v}$ is said to be irrotational if $\text{curl}(\bar{v})=0$.

$$\text{curl } \bar{v} = \nabla \times \bar{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ e^{xy}y & e^{xy}x & 2e^z \end{vmatrix} = \hat{i}(\frac{\partial}{\partial y} 2e^z - \frac{\partial}{\partial z} e^{xy}x)$$

$$- \hat{j}(\frac{\partial}{\partial x} 2e^z - \frac{\partial}{\partial z} e^{xy}y) + \hat{k}(\frac{\partial}{\partial x} e^{xy}x - \frac{\partial}{\partial y} e^{xy}y)$$

$$= \hat{i}.0 - \hat{j}.0 + \hat{k}(e^{xy} + e^{xy}yx - e^{xy} - e^{xy}xy) = 0$$

Example 6. If the vector field defined by the vector function $\vec{F} = xyz(yz\hat{i} + xz\hat{j} + xy\hat{k})$ is conservative then find the scalar potential function f .

Solution: If the given vector field is conservative then it can be expressed as the gradient of a scalar function $f(x, y, z)$. i.e. $\vec{F} = \text{grad}(f)$. Hence

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \vec{v} = xyz(yz\hat{i} + xz\hat{j} + xy\hat{k})$$

Comparing, we get

$$\frac{\partial f}{\partial x} = xy^2z^2, \frac{\partial f}{\partial y} = x^2yz^2, \frac{\partial f}{\partial z} = x^2y^2z.$$

Integrating the first equation, we get $f(x, y, z) = \frac{1}{2}x^2y^2z^2 + g(y, z)$.

Substituting in the second equation, we get

$$\frac{\partial f}{\partial y} = x^2 y z^2 = x^2 y z^2 + \frac{\partial g}{\partial y} \text{ or } \frac{\partial g}{\partial y} = 0 \text{ or } g = g(z).$$

Substituting in the third equation, we get

$$\frac{\partial f}{\partial z} = x^2 y^2 z = x^2 y^2 z + \frac{dg}{dz} \text{ or } \frac{dg}{dz} = 0 \text{ or } g = c, \text{ constant.}$$

Hence, $f(x, y, z) = \frac{1}{2} x^2 y^2 z^2 + c$. Therefore, there exist a scalar function $f(x, y, z)$ such that $\nabla f = \bar{F}$ and the vector field is conservative.

Assignment

(i) Find the $\text{div}(\bar{v})$, $\text{curl}(\bar{v})$ and verify that $\text{div}(\text{curl}(\bar{v})) = 0$ for

(a) $\bar{v} = (x\hat{i} + y\hat{j} + z\hat{k})$

(b) $\bar{v} = (x^2\hat{i} + y^2\hat{j} + z^2\hat{k})$

Thank you